

ORGANIC HEMP MATERIALS SCIENCE

Epoxidized Hempseed Oil (Hempoxies), High-Tenacity Bast Reinforcements & Vitrimeric Composites

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Scope: Industrial Hemp Mechanics, Epoxidation Lipochemistry, Fiber Interfacial Coupling, Structural Bio-Resins

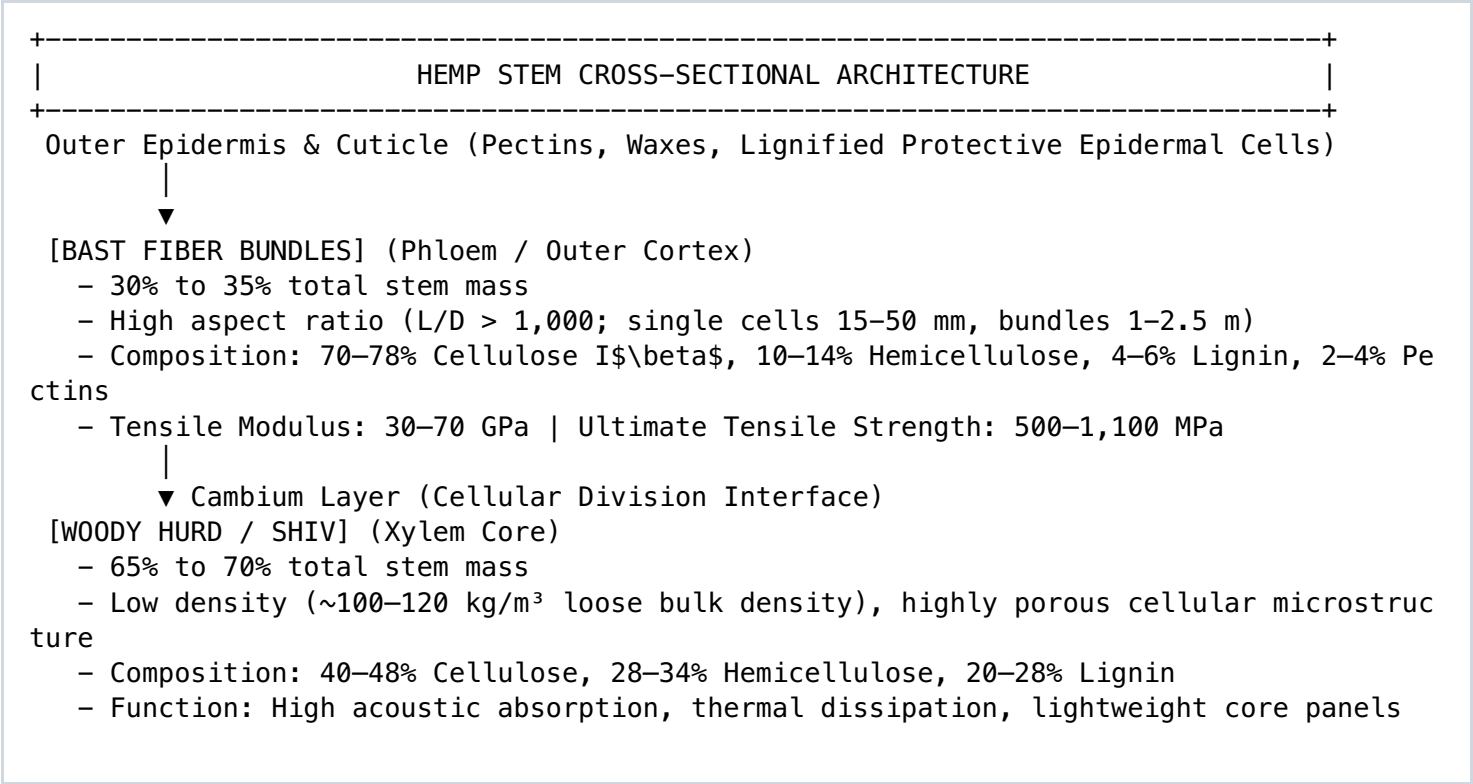
ENGINEERING DIRECTIVE: Modern structural composites face an unsustainable dependency on petrochemical diglycidyl ether of bisphenol-A (DGEBA) epoxy matrices and non-recyclable synthetic glass or carbon fibers. *Cannabis sativa L.* (industrial hemp) resolves both sides of the composite equation simultaneously: its high-aspect-ratio bast fibers provide specific tensile properties rivaling E-glass, while the polyunsaturated triglyceride profile of hempseed oil provides the chemical backbone for **Hempoxies**—fully renewable, epoxidized bio-thermoset and vitrimeric resins. This manual establishes the chemical synthesis, fiber-surface metallurgy, interfacial shear mechanics, and vacuum infusion processing required to manufacture structural-grade, carbon-negative composite architectures.

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1. PLANT ANATOMY & MACROMOLECULAR PHYSICS: BAST VS. HURD

The industrial hemp stalk (*Cannabis sativa L.*) is a naturally engineered, anisotropic core-shell biocomposite structure. It comprises two mechanically distinct cellulosic fractions possessing diametrically opposed macromolecular architectures:



Macromolecular Alignment & Microfibrillar Angle (MFA)

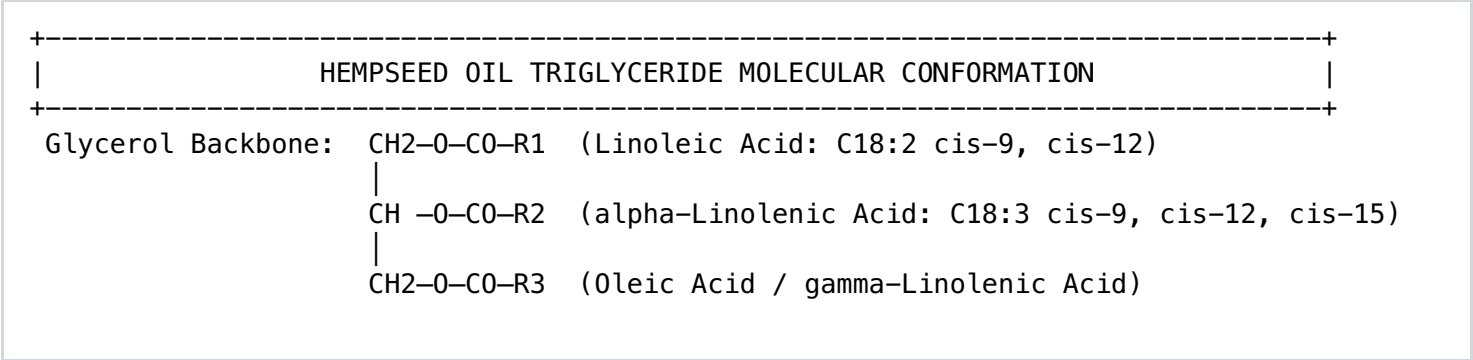
The exceptional axial tensile stiffness of hemp bast fibers is governed by their crystalline cellulose microfibrillar angle (θ_{MFA}). Inside the secondary cell wall (S_2 layer, which accounts for $>80\%$ of total fiber volume), cellulose chains form monoclinic $I\beta$ lattices oriented at a narrow helical angle of 2° to 6° relative to the longitudinal fiber axis.

In contrast to cotton ($\theta_{\text{MFA}} \approx 20^\circ\text{--}40^\circ$) or coir ($\theta_{\text{MFA}} \approx 30^\circ\text{--}50^\circ$), hemp's almost parallel microfibrillar orientation eliminates spiral unwinding under mechanical tension, generating a high specific modulus that directly matches or exceeds commercial E-glass fibers:

Structural Reinforcement Fiber	Density (ρ , g/cm³)	Tensile Strength (σ_t , MPa)	Young's Modulus (E, GPa)	Specific Modulus (E / ρ)
Industrial Hemp Bast (Technical)	1.45 – 1.48	550 – 1,100	40 – 70	27.5 – 47.3
Standard E-Glass Fiber	2.54 – 2.60	2,000 – 3,400	72 – 76	28.3 – 29.9
Flax Bast Fiber	1.45 – 1.50	600 – 1,000	50 – 80	34.4 – 53.3
Polypropylene (Matrix Polymer)	0.90 – 0.92	30 – 40	1.2 – 1.6	1.3 – 1.7

2. HEMPSEED OIL LIPOCHEMISTRY: PUFA DISTRIBUTIONS

Epoxidized vegetable oils (EVOs) depend strictly on the presence of aliphatic $C=C$ double bonds within their fatty acid acyl chains. Hempseed oil (*Cannabis sativa* L.* seed pressing) possesses one of the highest concentrations of **polyunsaturated fatty acids (PUFAs)** of any industrial oilseed crop ($>75\text{--}80\%$ total unsaturated fraction), characterized by an Iodine Value (IV) ranging between **140 and 170 g I_2 / 100g**.



Fatty Acid Chain Name	Lipid Shorthand	Mass Fraction Range (%)	Number of Double Bonds (\$C=C\$)	Target Oxirane Equivalent Yield
Linoleic Acid	C18:2 (\$\Delta^{9,12}\$)	52.0% – 60.0%	2 (\$C=C\$)	Di-epoxide precursor
α -Linolenic Acid (ALA)	C18:3 (\$\Delta^{9,12,15}\$)	15.0% – 25.0%	3 (\$C=C\$)	Tri-epoxide precursor (High crosslink density)
Oleic Acid	C18:1 (\$\Delta^9\$)	8.0% – 13.0%	1 (\$C=C\$)	Mono-epoxide chain extender
γ -Linolenic Acid (GLA)	C18:3 (\$\Delta^{6,9,12}\$)	2.0% – 4.0%	3 (\$C=C\$)	Tri-epoxide precursor
Saturated (Palmitic/Stearic)	C16:0 / C18:0	8.0% – 12.0%	0 (\$C=C\$)	Non-reactive dangling chain segments

The Theoretical Maximum Oxirane Oxygen Content (\$OOC_{\text{theor}}\$)

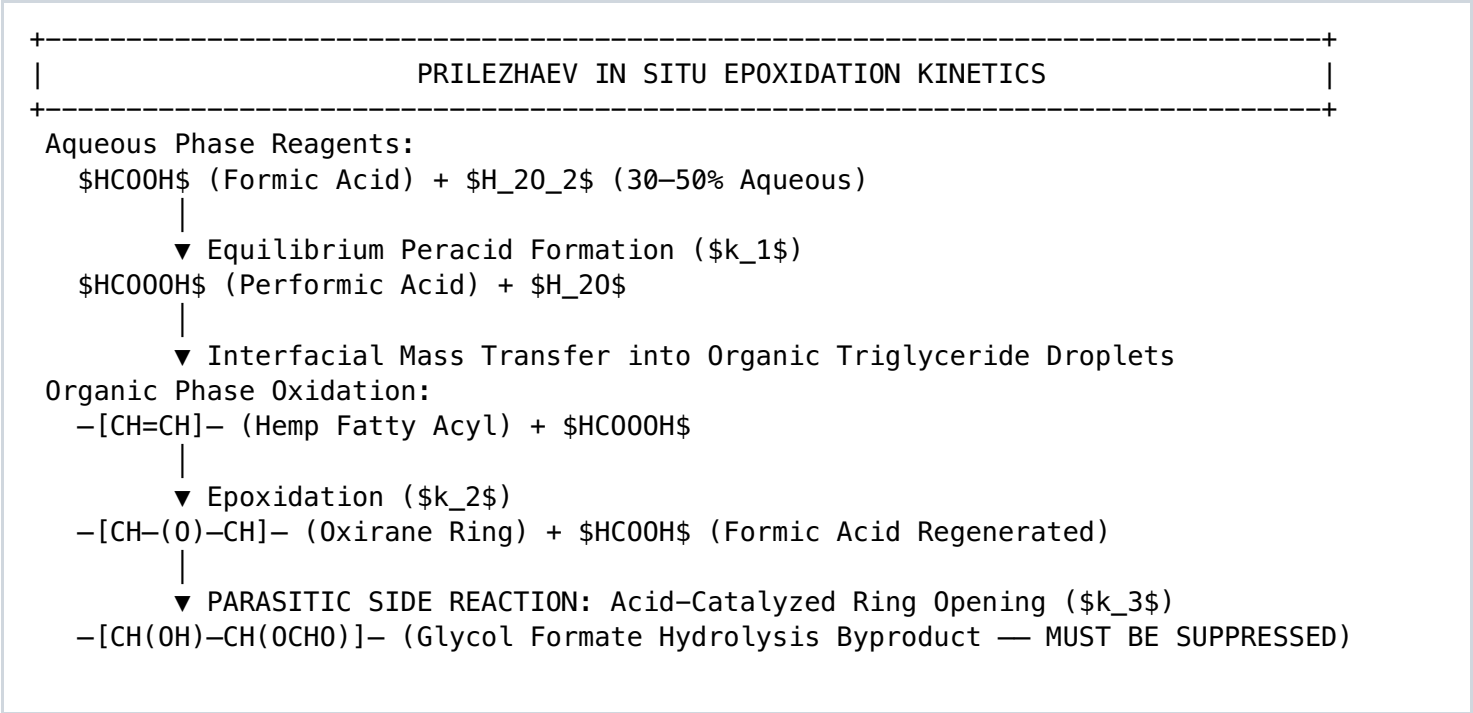
The concentration of epoxide groups achievable during synthesis scales directly with the initial Iodine Value (\$IV_0\$). The theoretical maximum weight percentage of oxirane oxygen (\$OOC_{\text{theor}}\$) is calculated via stoichiometry:

$$OOC_{\text{theor}} = [(IV_0 / (2 * 126.9)) * 16.00] / [100 + (IV_0 / (2 * 126.9)) * 16.00] * 100\%$$

For refined industrial hempseed oil with an initial $IV_0 = 165 \text{ g I}_2 / 100 \text{ g}$, the theoretical maximum oxirane oxygen is **\$9.43 \text{ wt}\%**. This high oxirane density provides Hempoxies with significantly higher crosslinking capability and elevated glass transition temperatures (\$T_g\$) compared to epoxidized soybean oil (ESBO, \$OOC \approx 6.5\text{--}7.0\%).

3. THE SYNTHESIS OF HEMPOXIES: KINETICS & OXIRANE FUNCTIONALITY

Epoxidizing hempseed oil converts localized aliphatic olefinic bonds ($C=C$) into three-membered oxirane (epoxy) rings. Industrial scale synthesis utilizes an *in situ* Prilezhaev reaction: a short-chain carboxylic acid (formic acid, $HCOOH$, or glacial acetic acid, CH_3COOH) acts as an oxygen carrier, reacting with hydrogen peroxide (H_2O_2) in the presence of an inorganic or heterogeneous acid catalyst:



Controlling Side Reactions: Temperature & Degradation Kinetics

The epoxidation reaction is highly exothermic ($\Delta H_{rxn} \approx -230 \text{ to } -250 \text{ kJ per mole of } C=C$). The primary competitive reaction is the premature, acid-catalyzed hydrolysis of the oxirane ring into vicinal diols, hydroxy-esters, and polymeric polyols.

- Thermal Optimization:** Epoxidation must proceed within a strict temperature window: **50°C to 65°C**. Below 45°C, peracid generation kinetics are slow; above 70°C, the rate of oxirane cleavage (k_3) exceeds the rate of epoxidation (k_2), reducing the final oxirane oxygen content.
- Heterogeneous Catalysis:** To eliminate mineral acid-induced ring opening, modern synthesis utilizes acidic ion-exchange resins (e.g., Amberlite IR-120 or Dowex 50WX2). The macroporous sulfonic acid beads remain solid, catalyzing peracetic acid formation while remaining physically excluded from the hydrophobic oil droplet interior, minimizing internal oxirane hydrolysis.

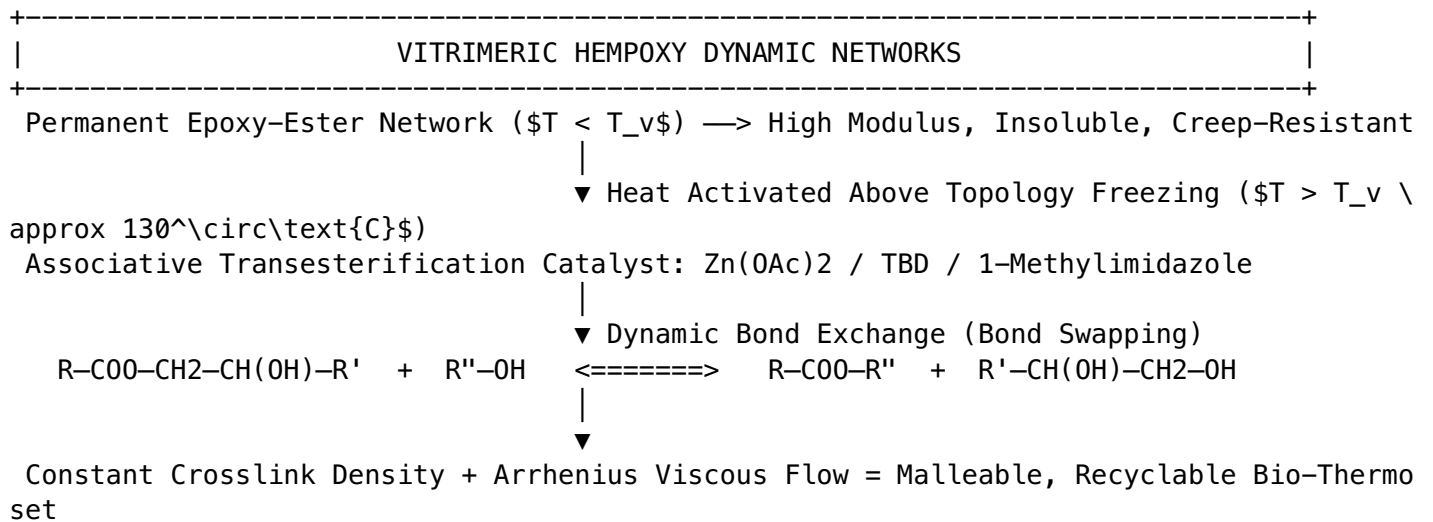
Operational Variable	Optimal Synthesis Parameter Range	Chemical Consequence of Deviation
Molar Ratio (\$C=C : HCOOH : H_2O_2\$)	1.00 : 0.25–0.50 : 1.50–1.80	Deficiency halts conversion; excess \$H_2O_2\$ accelerates ring opening.
Addition Rate of \$H_2O_2\$	Controlled dropwise across 90–120 min	Rapid addition causes localized thermal runaways and rapid foaming.
Stirring Speed / Agitation	800 – 1,200 RPM (High Shear)	Overcomes liquid-liquid mass transfer boundaries between water and oil.
Final Epoxy Equivalent Weight (EEW)	180 – 210 g / eq	Target industrial range for high-crosslink composite matrices.

4. CROSSLINKING DYNAMICS: HARDENERS & VITRIMERIC TOPOLOGY

Uncured epoxidized hempseed oil (EHO) is a viscous amber fluid ($\eta \approx 300\text{--}600\text{ mPa}\cdot\text{s}$ at 25°C). Transforming EHO into a rigid structural thermoset requires crosslinking with polyfunctional co-monomers. Because of steric hindrance surrounding the internal oxirane rings along the fatty acid chains, EHO exhibits slower curing kinetics than terminal-glycidyl petrochemical epoxies like DGEBA.

Crosslinking Hardener Chemistries

- 1. Cycloaliphatic & Bio-Anhydrides:** Systems cured with Methylhexahydrophthalic anhydride (MHHPA) or bio-derived Succinic/Citric anhydrides yield rigid ester crosslinks with high optical clarity, superior electrical resistivity, and high thermal decomposition thresholds ($T_d > 320^\circ\text{C}$). The reaction proceeds via alternating polyesterification catalyzed by tertiary amines (e.g., 2-ethyl-4-methylimidazole, 2-EMI).
- 2. Bio-Based Phenalkamines & Dimer Diamines:** Derived from cashew nutshell liquid (CNSL) or dimerized vegetable fatty acids (e.g., Priamine 1074). They permit room-temperature ambient curing while imparting flexible impact absorption and hydrophobic moisture resistance to the cured matrix.

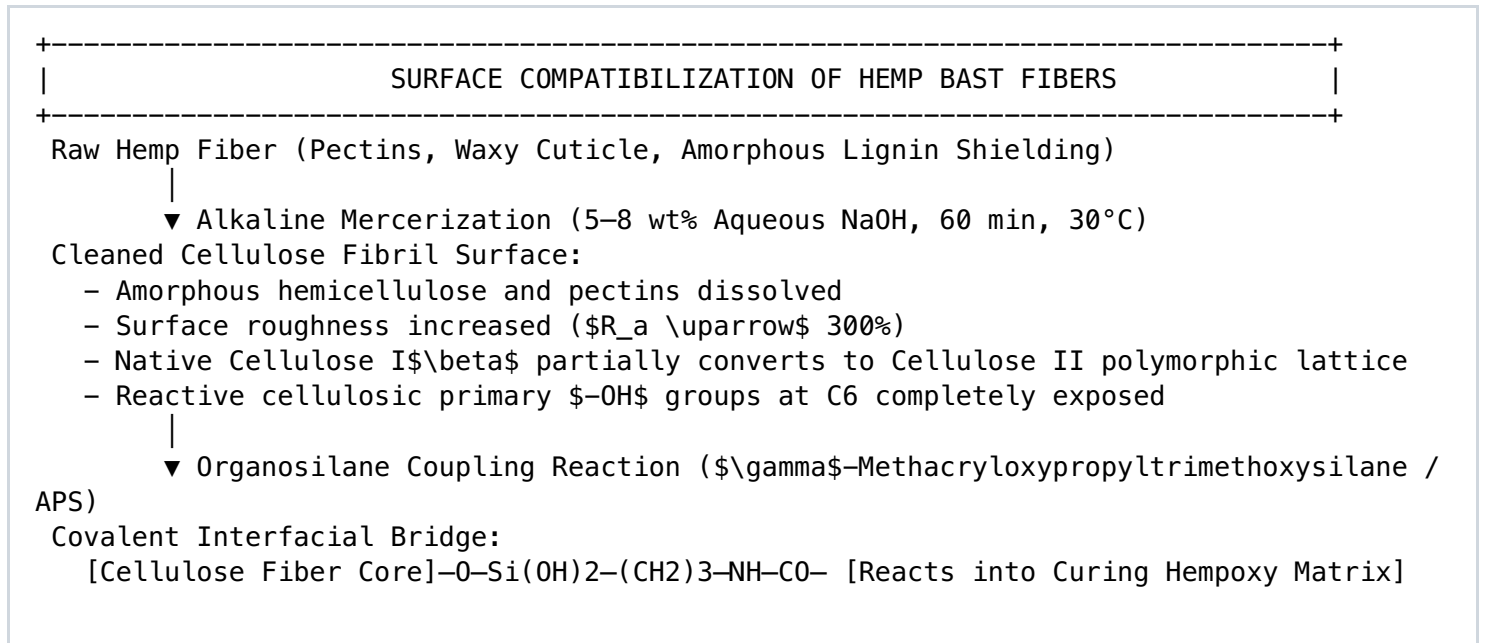


The Vitrimeric Solution: Eliminating Composite Waste

Conventional epoxies cannot be melted or recycled; components must be landfilled or pyrolyzed at end-of-life. By engineering Hempoxies as **covalent adaptable networks (vitrimers)** through catalytic zinc transesterification, the crosslinked hemp composite can be ground, re-formed under hot-press thermo-compression at 160°C , and fully recovered without polymer matrix degradation.

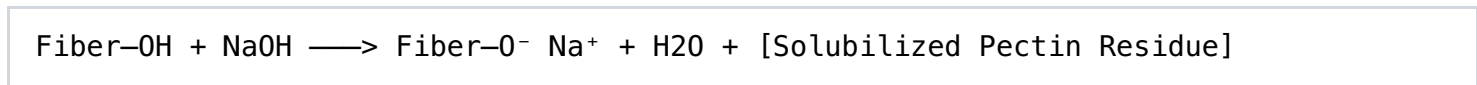
5. INTERFACIAL ENGINEERING: MERCERIZATION & SILANE COUPLING

The primary failure mode in natural fiber-reinforced composites is interfacial shear failure occurring at the boundary between the **hydrophilic cellulose fiber** (rich in surface hydroxyl groups, $-OH$) and the **hydrophobic polymer matrix**. Without chemical modification, interfacial voids accelerate microcracking and moisture wicking.



Alkaline Mercerization Mechanics

Soaking raw unspun hemp bast bundles in an aqueous sodium hydroxide (NaOH) solution strips the amorphous cementing compounds (hemicellulose, pectins, surface waxes):

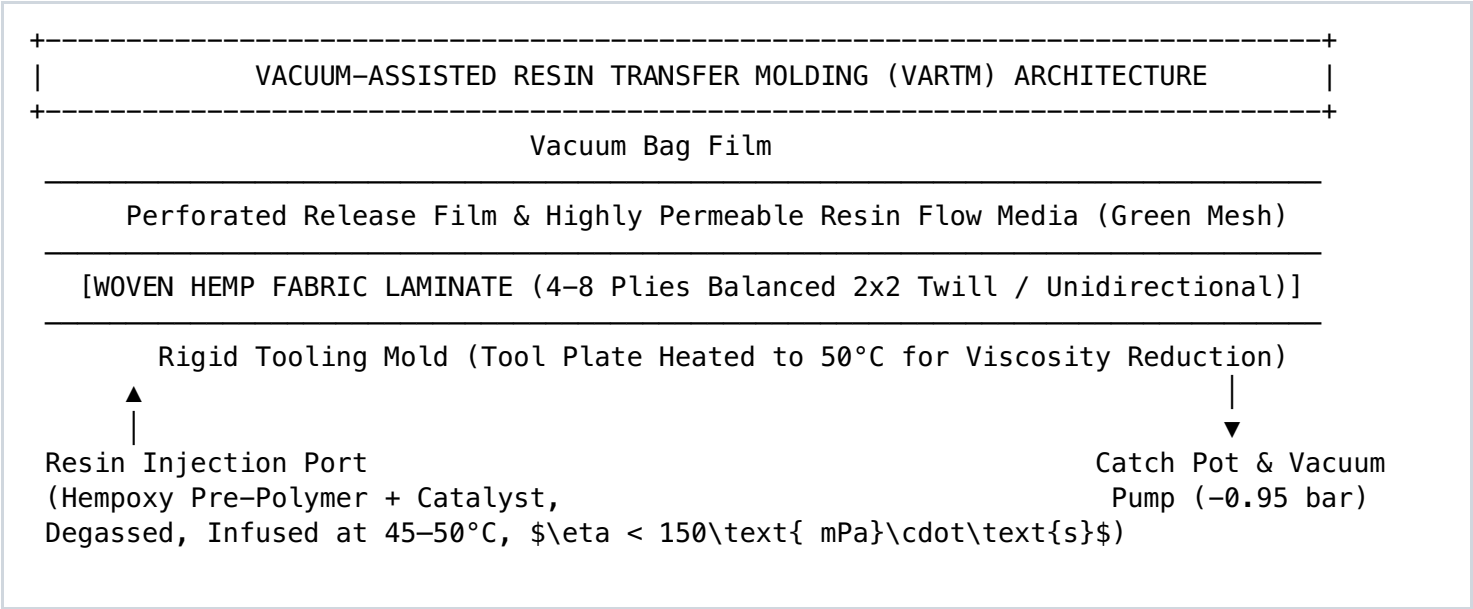


Excessive alkaline exposure ($>10\text{ wt\%}$ NaOH or temperatures $>60^{\circ}\text{C}$) degrades crystalline cellulose chains through alkaline peeling reactions, significantly reducing absolute fiber tensile strength. The process window must be maintained at **5.0 to 6.0 wt% NaOH** at ambient temperatures for precisely **45 to 60 minutes**.

Fiber Treatment Regime	Surface Energy (\$\gamma_s\$, mN/m)	Interfacial Shear Strength (IFSS, MPa)	Composite Tensile Modulus (\$E\$, GPa)
Untreated Raw Bast Fiber	32.4 (Weak Waxy Wetting)	8.2 ± 1.1	18.4 ± 1.2
Alkalized (6 wt% NaOH)	48.6 (High Roughness)	16.5 ± 1.4	24.8 ± 1.5
Alkalized + 1.0% APTES Silane	58.2 (Covalently Reactive)	26.4 ± 1.8	32.6 ± 1.4
Enzymatic Retting (Pectinase)	44.1 (Clean Non-Damaged)	14.8 ± 1.2	22.1 ± 1.3

6. ADVANCED COMPOSITE MANUFACTURING: VARTM & PRE-PREGS

Manufacturing structural hempoxy composites requires specific processing controls. Unlike synthetic glass or carbon reinforcements, natural cellulosic fibers are porous, compressible, and moisture-sensitive, requiring precise control over resin viscosity, vacuum limits, and cure schedules.



The Moisture Volatilization Boundary

Natural hemp fibers are intrinsically hygroscopic, carrying between 8% and 12% adsorbed water under ambient storage. **Direct processing without rigorous pre-drying causes structural composite failure:** at curing temperatures exceeding 100°C, entrapped moisture

vaporizes into steam, generating internal interlaminar voids, blistering, and incomplete cure stoichiometry due to water-induced anhydride hydrolysis.

- **Thermal Dehydration Protocol:** Prior to mold layup, all woven or non-woven hemp reinforcements must be baked in a vented air-circulation oven at **105°C for a minimum of 4 hours** (or vacuum-dried at 80°C at -0.09 MPa for 2 hours) to bring total moisture content below **0.5 wt%**.
- **Processing Viscosity Window:** Uncut EHO has a viscosity of ~400 mPa·s at 25°C. To ensure complete wetting of dense bast bundles during VARTM infusion, the resin must be pre-heated to **45°C–50°C**, dropping the injection viscosity below **120 mPa·s** before entering the dry reinforcement pack.

7. MECHANICAL & DYNAMIC PERFORMANCE OF HEMPOXY LAMINATES

When fully cured, hempoxy-bast laminates provide specific mechanical performance competitive with synthetic fiberglass-reinforced plastics (GFRP), accompanied by improved acoustic dampening and non-catastrophic failure mechanisms.

Mechanical Metric (ASTM Standard)	Neat Cast Hempoxy Resin	Woven Hemp Bast / Hempoxy Laminate	Commercial E-Glass / Petrochemical Epoxy
Tensile Strength (σ_t , MPa)	42 – 55	180 – 260	320 – 450
Tensile Modulus (E_t , GPa)	1.8 – 2.4	16.5 – 24.5	20.0 – 26.0
Flexural Strength (σ_f , MPa)	65 – 82	195 – 240	380 – 500
Flexural Modulus (E_f , GPa)	2.1 – 2.6	14.0 – 19.5	18.0 – 22.0
Izod Notched Impact (kJ/m^2)	3.2 – 4.5	32.0 – 45.0	50.0 – 80.0
Glass Transition (T_g via DMA $\tan \delta$)	85°C – 125°C	95°C – 135°C	120°C – 160°C
Loss Factor (Damping Coefficient, $\tan \delta$)	0.02	0.065 (High Acoustic Absorption)	0.012 (Prone to Resonant Ringing)

Impact Dissipation & Pseudo-Ductility

Synthetic glass and carbon laminates fail under ballistic impact via abrupt, brittle fracture propagation. Hemp bast fibers, owing to their concentric multilayered cell wall morphology (S_1 , S_2 , S_3 secondary wall lamellae) and lumen core, deform through controlled micro-delamination, fiber pull-out, and progressive cell-wall shearing. This pseudo-ductile failure mechanism absorbs significant kinetic energy, making hempoxy plates suitable for blast attenuation, automotive crash boxes, and protective tactical shells.

8. HYGROTHERMAL AGING & FLAME RETARDANCY METRICS

Fickian Moisture Diffusion Kinetics

Moisture ingress in hempoxy composites follows classic Fickian diffusion models across initial exposure periods, transitioning to non-Fickian matrix swelling under prolonged saturation:

$$M_t / M_{\infty} = \frac{4}{h} \cdot \sqrt{\frac{D \cdot t}{\pi}}$$

Where M_t is moisture uptake at time t , M_{∞} is saturation uptake, h is specimen thickness, and D is the diffusion coefficient. Neat epoxidized hemp oil matrices possess aliphatic fatty chains that act as hydrophobic barriers, reducing the diffusion coefficient ($D \approx 1.2 \times 10^{-6} \text{ mm}^2/\text{s}$) compared to conventional polar amine-cured DGEBA matrices.

Flame Retardancy: Intumescent Mineral Functionalization

Cellulose and bio-resins are naturally combustible organic structures. To achieve UL94 V-0 flame ratings without using toxic halogenated flame retardants, the matrix is compounded with bio-based phosphorus-nitrogen additives:

- **Phytic Acid (Inositol Polyphosphate):** A renewable plant extract containing six phosphate ester groups per ring. When compounded into Hempoxies with ammonium polyphosphate (APP) or melamine, it acts as a catalytic acid source, triggering rapid surface carbonization (intumescent char formation) upon flame contact.
- **Char Barrier Mechanism:** The generated phosphorus-carbon char crust acts as an insulating barrier with high thermal impedance, blocking oxygen access to the inner composite plies and increasing the Limiting Oxygen Index (LOI) from 19.5% to over **31.0%**.

9. MASTER LABORATORY FORMULATIONS & PROTOCOLS

Protocol 01: Pilot-Scale Synthesis of Epoxidized Hempseed Oil (EHO / Hempoxy)

Landry Industries
Specification EH0-Rev4

Reaction Stoichiometry:

- Virgin Cold-Pressed Hempseed Oil (IV = 165 g I2/100g): 500.0 g (~3.25 mol C=C double bonds).
- Formic Acid (HCOOH, 98% ACS Grade): 37.4 g (0.81 mol; 0.25 molar equivalent relative to C=C).
- Hydrogen Peroxide (H2O2, 50 wt% Aqueous Solution): 331.5 g (4.88 mol; 1.50 molar equivalent).
- Toluene / CPME (Optional organic carrier solvent): Solventless preferred.

Reactor Configuration:

- 2,000 mL five-necked round-bottom borosilicate reactor equipped wi